

Kobe Guo

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science in Data Science (MCS-DS)

Champaign, IL

Expected May 2027

- **GPA: 4.0**

- Relevant coursework: Applied Machine Learning, Data Mining, Cloud Computing, Deep Learning

Vanderbilt University

Bachelor of Science in Computer Science, Minor in Data Science

Nashville, TN

Aug. 2020 – May 2024

- **GPA: 3.42**

- Relevant Coursework: Algorithms, Data Structures, Database Management Systems, Machine Learning

EXPERIENCE

Software Engineer Intern

June 2026 – Aug. 2026

Capital One

McLean, VA

- Built native Android features for Capital One's mobile banking app using **Kotlin** and **Jetpack Compose**, reducing screen load time by **22%** through optimized state management with **MVVM** and **Kotlin Coroutines**.
- Developed and tested **RESTful API** integrations via **Retrofit**, increasing unit test coverage by **18%** and eliminating **2** critical bugs caught pre-release across **65M+** customer-facing flows.

Software Engineer Intern

June 2023 – Aug. 2023

Nourish'd

Cape Town, South Africa

- Developed a React Native app for food ordering, **increasing user engagement by 30%**, with a **50% rise in daily active users**.
- Migrated backend from **MySQL to Firebase**, improving scalability and reducing data retrieval times through real-time synchronization, **boosting backend efficiency by 25%**.

PROJECTS

CaliForest (PyHealth Open-Source Contribution) | *Python, scikit-learn, NumPy, PyTorch* [Github](#) | [PR #999](#)

- Implemented and contributed a **calibrated Random Forest (CaliForest)** model to the PyHealth open-source library; PR reviewed and **successfully merged**.
- Designed calibration pipeline using **out-of-bag predictions and tree-level variance** with inverse-variance weighting; supported **isotonic and logistic regression**.
- Refactored training to a dedicated `fit()` pipeline (vs minibatch forward pass) and validated on **MIMIC-IV ICU data** using **AUROC and Brier score**.

URL Shortener | *Node.js, Express, Redis, PostgreSQL, React, Docker, AWS EC2*

[Github](#) | [Live Demo](#)

- Built a production URL shortener implementing **cache-aside pattern** with Redis and PostgreSQL, achieving **5,998 req/sec at 50ms p99 latency** under 100 concurrent connections.
- Implemented **Redis-backed rate limiting** (10 req/min per IP) and async click analytics tracking user-agent, IP, and timestamp per redirect.
- Deployed to **AWS EC2** with Docker Compose, pm2 process management, automated startup migrations, and a React/Tailwind frontend served in production.

Medical Image Segmentation with Deep Learning | *Python, TensorFlow, Keras*

[Github](#) | [View Project](#)

- Trained a U-Net model on the ULS23 challenge dataset of 3D medical images sourced from hospitals, achieving a **0.87 Dice score** on bone datasets, up from a 0.75 baseline.
- Addressed 3D volume memory constraints by slicing volumetric scans into 2D cross-sections, enabling training on full hospital-grade imaging data without GPU memory overflow.
- Improved model robustness through data augmentation including random flipping, rotation, and noise injection across a multi-contributor medical imaging dataset.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, SQL

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Redux, TensorFlow, PyTorch, scikit-learn

Databases & Caching: PostgreSQL, MySQL, Redis, Firebase

Cloud & DevOps: AWS EC2, Docker, Docker Compose, CI/CD